

HEMANTH AKKENAPALLY

RF & Biomedical Systems Engineer | Data Scientist | Machine Learning and Signal Analysis

Baltimore, MD | (667) 406-4046 | akkenapallyhemanth@gmail.com | hemanthakkenapally.com | LinkedIn | GitHub

SUMMARY Interdisciplinary engineer and data scientist with an MPS in Data Science, a B.Tech in Electronics and Communication Engineering, and experience spanning biomedical catheter R&D, RF simulation and measurement, signal analysis, machine learning, enterprise data engineering, and network systems. Research interests center on RF and microwave systems, biomedical sensing, scientific machine learning, experimental data analysis, and reliable intelligent systems.

EDUCATION

Master of Professional Studies in Data Science | University of Maryland, Baltimore County (UMBC) | Baltimore, MD

Dec 2024

GPA: 3.74/4.00.

Bachelor of Technology in Electronics and Communication Engineering | Sreenidhi Institute of Science and Technology |

Hyderabad, India

Jul 2021

CGPA: 8.86/10.

EXPERIENCE

Systems Engineer, Cardiac Catheter R&D | SIGT LLC | Baltimore, MD

Jan 2025 - Present

- Design catheter electrodes and related assemblies in Ansys HFSS; evaluate simulated electromagnetic behavior before designs are released for manufacturing.
- Use vector network analyzer measurements and S-parameters to characterize RF response across catheter configurations, input powers, and frequencies.
- Design and apply ninth-order low-pass filters for RF signal analysis and integrate VNA, oscilloscope, catheter telemetry, and experimental metadata for analysis.
- Investigate relationships among measured RF response, cardiac-tissue lesion depth, and microbubble formation to support experimental interpretation and device development.
- Build Python and SQL pipelines, dashboards, statistical analyses, and machine-learning workflows for research data quality, visualization, and predictive modeling.

Data Engineer & Salesforce Developer | Cognizant Technology Solutions | India

Sep 2021 - Jan 2023

- Developed and maintained Salesforce, Apex, SOQL, SQL, Snowflake, ETL, data-modeling, and reporting workflows for financial and loan-management processes.
- Implemented validation, access controls, audit-oriented data practices, dashboards, and CI/CD-supported releases in cross-functional Agile environments.
- Translated operational requirements into data models, automated workflows, analytical queries, and stakeholder-facing reports.

Network Engineering Trainee / Internship | Cisco Networking Academy-oriented Program | India

May 2019 - May 2021

- Completed Introduction to Networks and Switching, Routing, and Wireless Essentials training with Packet Tracer-based routing and switching exercises.
- Applied OSI and TCP/IP concepts, network telemetry, SQL, ETL, and dashboarding to operational monitoring and performance-analysis exercises.

PEER-REVIEWED PUBLICATIONS

1. Pendli Pradeep, Hemanth Akkenapally, Gundu Kavya Sri, and Sama Vinisha. "A Compact Frequency Reconfigurable Antenna with Enhanced Bandwidth for WLAN Applications." International Journal of Creative Research Thoughts, vol. 9, no. 6, pp. c6-c12, June 2021.
 - HFSS simulation of a compact FR-4 tuning-fork antenna with partial ground and PIN-diode models for switchable WLAN/C-band resonances.
2. Pendli Pradeep, Hemanth Akkenapally, Gundu Kavya Sri, and Sama Vinisha. "A Compact Metamaterial Based Dual-Band Antenna with Improved Gain for WLAN Applications." International Journal for Research in Applied Science & Engineering Technology, vol. 9, no. VI, pp. 2778-2783, June 2021. DOI: 10.22214/ijraset.2021.35586.
 - HFSS simulation of a compact FR-4 split-ring-resonator antenna with defected ground and MATLAB-supported parameter extraction for metamaterial characterization.

RESEARCH & ACADEMIC PROJECTS

Compact and multiband antenna research: Designed and analyzed frequency-reconfigurable, metamaterial dual-band, and multi-slot microstrip patch antenna concepts using HFSS and RF design principles.

TweetSense: Built a real-time sentiment-analysis pipeline using BERT, PyTorch, AWS S3/Lambda, and a REST API for large-scale social-media text processing.

Automatic license-plate recognition: Developed a YOLOv4/Darknet/OpenCV pipeline with OCR and Python for vehicle plate detection and character recognition.

Online doctor appointment application: Built a Java web application using servlets, HTML, CSS, JavaScript, and SQL for appointment scheduling and database-backed workflows.

Salesforce loan-management application: Created an application supporting loan application and management workflows using Salesforce administration and development tools.

SKILLS

RF, electronics & experimental systems: Ansys HFSS, catheter electrode simulation, VNA measurements, S-parameters, RF filtering, antennas, microwave systems, DSP, analog/digital communications, VLSI, FPGA/Verilog, and experimental signal analysis.

Programming & scientific computing: Python, SQL, Pandas, NumPy, Spark/SparkSQL, R, SAS, Bash, C/C++, Java, JavaScript, HTML, and CSS.

Machine learning & AI: scikit-learn, TensorFlow, PyTorch, statistical modeling, regression/classification, NLP, computer vision, feature engineering, model evaluation, and time-series analysis.

Data engineering: ETL/ELT, Airflow, Kafka, Snowflake, Redshift, data lakes, data warehouses, data modeling, data quality, metadata, and time-series data optimization.

Cloud & platforms: AWS S3, Glue, Lambda, EMR, Redshift, SageMaker, EC2, IAM, and KMS; Azure Data Factory, Data Lake, and Synapse; GCP BigQuery.

Software & analytics: Flask, FastAPI, REST APIs, Docker, Kubernetes, Git, CI/CD, Tableau, Power BI, Streamlit, Salesforce, Apex, and SOQL.

ACADEMIC ENGAGEMENT & TRAINING

- Participated in a machine-learning workshop at the Indian Institute of Technology Hyderabad, including Python-based classification and computer-vision exercises.
- Participated in a national-level technical project expo with an agriculture-portal project.
- Completed Cisco networking coursework in Introduction to Networks and Switching, Routing, and Wireless Essentials, along with Python programming training.